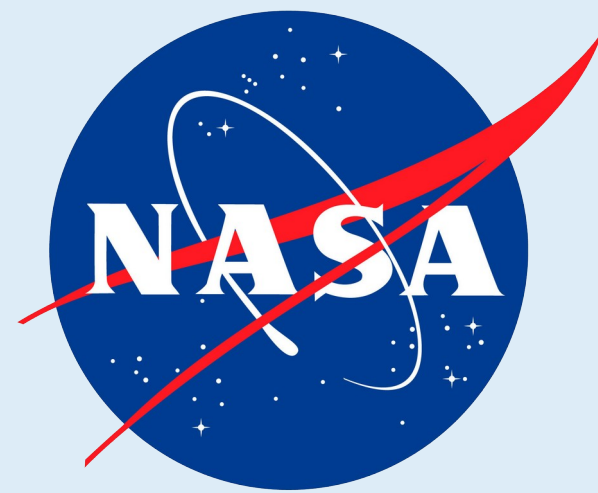




Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Spaceflight affects multiple organs, but differences among missions can obscure responses that recur across studies. We asked whether pathway-constrained expiMap models could identify reproducible programs in NASA OSDR mouse bulk RNA-seq. Samples were mapped into tissue-matched ARCHS4 references using approximately 2,000 highly variable genes linked to current mouse Reactome pathways. Project-balanced flight-ground shifts were evaluated with enrichment, held-out projects, three complete trainings, composition proxies, and member-gene review. Retained patterns included lower repair and cytoskeletal programs in thymus, lower maintenance and barrier programs in skin, lower adaptive-immune programs in liver, and coordinated lower adaptive and innate immune programs in spleen, the strongest multi-pathway result.

INTRODUCTION

PROJECT OBJECTIVE

Identify gene programs that shift consistently across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression **Spaceflight biology** + **Tissue biology** + **Animal variation** + **Study / protocol**

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples. Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

ssGSEA + GSEA

Held-out projects

3 full trainings

Composition proxies

Member genes

Literature review

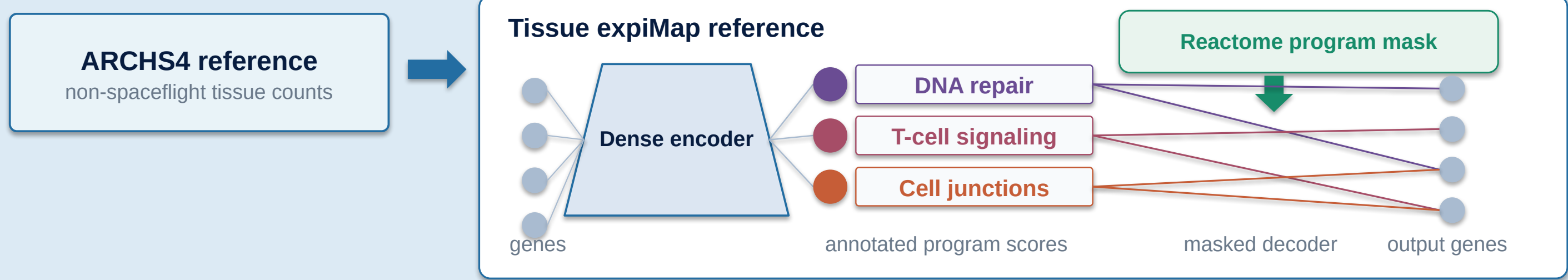
Complementary means an annotated Reactome program that adds a plausible perspective beyond the dominant phenotype in prior literature. De novo nodes were not retained in the final models.

METHODS

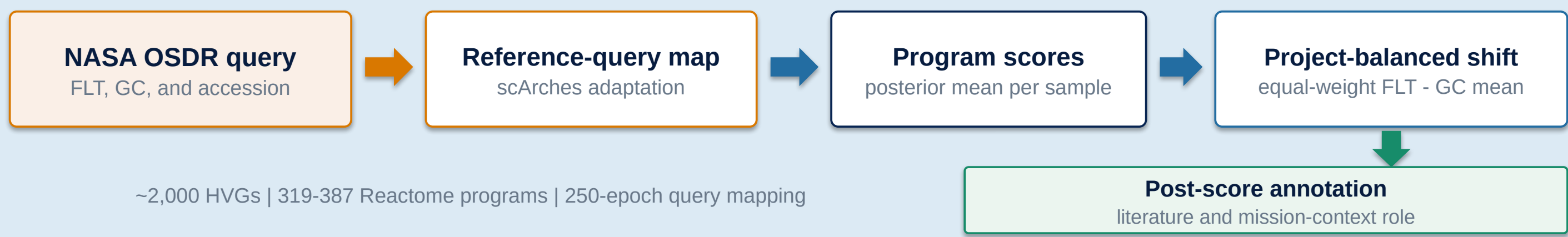
Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.

1 REFERENCE TRAINING

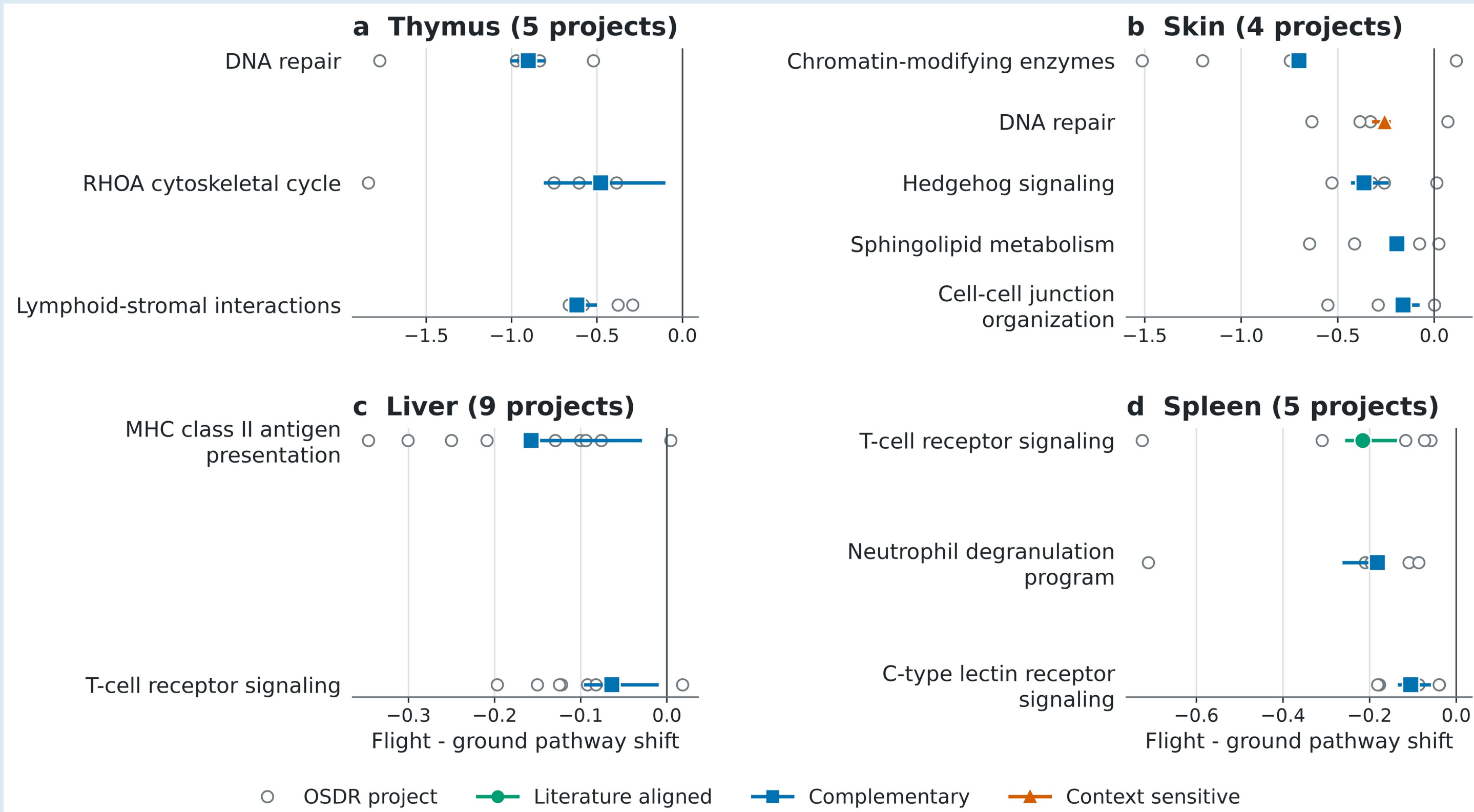


2 QUERY MAPPING AND CROSS-MISSION COMPARISON



~2,000 HVGs | 319-387 Reactome programs | 250-epoch query mapping

RESULTS: ALL 13 PRIMARY RETAINED PATHWAY SHIFTS



All 13 primary retained programs are shown; three exploratory kidney programs remain in the paper. Open circles are OSDR projects; colored ranges span three complete reference-query trainings. All displayed shifts are lower in flight; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Final robustness-filtered tissue stories

Flight-associated pathway-score patterns and tissue-state hypotheses

Tissue context	Observed lower expiMap program scores in flight	Interpretive link (not causal order)	Tissue-state hypothesis
A THYMUS 	↓ DNA repair ↓ RHOA cytoskeletal cycle ↓ Lymphoid-stromal interactions	----->	Consistent with reduced repair, cell motility, and niche coordination
B SKIN 	↓ Chromatin-modifying enzymes ↓ DNA repair ↓ Hedgehog signaling ↓ Sphingolipid metabolism ↓ Cell-cell junction organization	----->	Consistent with reduced tissue maintenance, barrier support, and cell coordination
C LIVER 	↓ MHC class II antigen presentation ↓ T-cell receptor signaling	----->	Consistent with lower adaptive immune communication
D SPLEEN 	↓ T-cell receptor signaling ↓ Neutrophil degranulation ↓ C-type lectin receptor signaling	----->	Consistent with lower adaptive activation, innate sensing, and effector-related transcription

Conceptual synthesis of bulk-tissue pathway scores. Lower scores do not prove pathway inhibition; cell-composition shifts may contribute.

CONCLUSIONS

- Spleen was strongest: T-cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue-maintenance programs; liver added a lower adaptive-immune axis to heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell-type-resolved pathway activity.

REFERENCES

1 Gebre et al., NAR 2025 (OSDR). 2 Lotfollahi et al., Nat Cell Biol 2023 (expiMap). 3 Lachmann et al., Nat Commun 2018 (ARCHS4). 4 Milacic et al., NAR 2024 (Reactome). 5 Horie et al., Sci Rep 2019. 6 Cope et al., Commun Med 2024. 7 Beheshti et al., Sci Rep 2019. 8 Gridley et al., J Appl Physiol 2009.

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github.com/jasont314/nasa-mouse